Design Limits

Tabular values in this section are to be used for design. Figures are provided for illustration only.

3.4.1 Lunar Surface Geological and Geomorphological Environment

Rationale: All lunar surface systems are influenced by lunar surface geology and geotechnical properties, so it is required for design of surface systems.

3.4.1.1 Crater Size-Frequency Distribution

The areal number density (number per unit area) of craters on the Moon is a function of surface age, as craters accumulate with time. For a surface of a given age, the crater density will follow the production function until the frequency crosses the equilibrium curve, then tracks the equilibrium density. As an illustration, for a 3.5 billion year old (Ga) surface typical of the lunar maria, the cumulative number of craters of diameter ≥*D* per km^2 can be summarized as:

Table 3.4.1.1-1. Number density (also called frequency) of craters equal to or larger than diameter D per km^2 for a 3.5 Ga surface.

D (km)	# ≥*D* (per km$^{2)}$	*D* (km)	# ≥*D* (per km$^{2)}$	*D* (km)	# ≥*D* (per km$^{2)}$
0.01	7.94E+02	0.1	7.94E+00	1	4.79E-03
0.0126	5.01E+02	0.126	5.01E+00	1.259	2.16E-03
0.0158	3.16E+02	0.158	3.16E+00	1.585	1.02E-03
0.0200	2.00E+02	0.200	1.72E+00	1.995	5.12E-04
0.0251	1.26E+02	0.251	7.73E-01	2.512	2.75E-04
0.0316	7.94E+01	0.316	3.39E-01	3.162	1.59E-04
0.0398	5.01E+01	0.398	1.46E-01	3.981	9.92E-05
0.0501	3.16E+01	0.501	6.17E-02	5.012	6.61E-05
0.0631	2.00E+01	0.631	2.60E-02	6.310	4.67E-05
0.0794	1.26E+01	0.794	1.10E-02	7.943	3.46E-05

Large craters here come from the production function of Neukum et al. (2001); surfaces older than 3.5 Ga will have accumulated more large craters than given in Table 3.4.1.1-1; younger surfaces will have fewer. Values for craters ≤200m in this Table follow

$$N_{equilibrium-Trask}(\geq D) = 0.079433D^{-2}$$

For sub-100m craters, this Trask equilibrium function is a valuable upper limit for the number of craters on a surface of any age. Although the Trask equation was based on craters D>10m, it is appropriate for design to extrapolate this equation to sizes smaller than 10m. Craters in this size range have reached equilibrium on every part of the Moon with the exception of very rare regions that were the site of large recent impacts (i.e., floors of large young craters). This equilibrium function is always an upper limit because surfaces with non-negligible topographic slope retain fewer craters. Figure 3.4.1.1-1 provides an illustration of the crater density in this table.

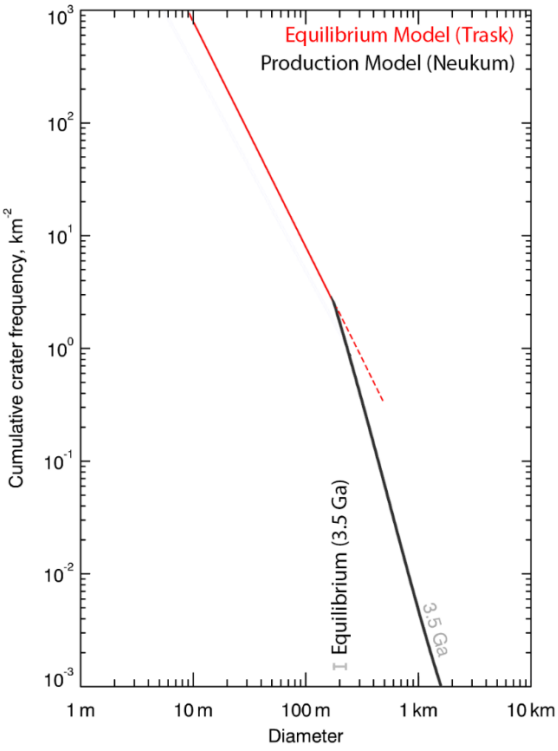


Figure 3.4.1.1-1. Cumulative crater density (number of craters of diameter ≥X per km^2) on a lunar surface of 3.5 Ga. (Ga=billions of years/surface age).

Design Limits

These series are generalizations that should be refined during detailed landing site characterization with direct observations of the crater population. The number density of craters can locally be highly affected by secondary craters and impact ejecta that will cause deviations in the number density of craters from the function. Secondary craters are usually recognizable from morphological considerations during detailed landing site characterization.

3.4.1.2 Crater Topography

Description

This section provides characteristics of individual craters' topography. See 3.4.1.3 for additional discussion on lunar topography and slope distributions.

Topography generated by craters provides most of the relief on the Moon. The topographic characteristics of individual craters vary strongly as a function of size and their age (as a result of the period of their exposure to the lunar surface environment). Larger craters remain minimally degraded for longer time periods. Morphologically fresh primary craters smaller than 10 km have floor-to-rim depth/diameter ratios given in Table 3.4.1.2-1.

Table 3.4.1.2-1. Depth/diameter ratios for fresh lunar craters (after Stopar et al., 2017).

Diameter Range	40-100m	100-200 m	200 m-400 m	400 m – 10 km
Fresh crater depth/diameter Ratio	0.12-0.14	0.14-0.16	0.16-0.17	0.19-0.22

Craters less than a few kilometers in size maintain their original depth for a geologically short period of time. For craters in the equilibrium part of the size distribution ($D \leq$100-m sizes, see 3.4.1.1), most craters are reduced to less than <50% of their initial depth. Characteristics of sub-km craters on a typical surface are:

Table 3.4.1.2-2. Crater morphological characteristics (for D<1 km; modified after Basilevsky, 1976).

	Freshest craters		→		**Degraded Craters**
Fraction of population	0.5%	2.5%	17%	30%	50%
Crater Characteristics	Very steep slopes, pronounced blocky ejecta, optically immature	Steep slopes, blocks common	Moderate slopes, most blocks on rim	Gentle slopes, rim mostly eroded	Very gentle slopes, no rim
Typical depth/diameter ratio	0.12-0.2+	0.12-0.2	0.1-0.15	0.07-0.1	<0.07
Max. Wall Slope	35˚+	25-35˚	15-25˚	10-15˚	<10˚

Because small craters degrade rapidly, most slope hazards at landing sites are from large craters ($D \geq$0.4 km) and/or fresh craters that are obvious during detailed landing site characterization.

Limitations

This description of craters' topography does not apply to large craters (≥10 km); craters larger than 15-20 km are in the complex crater regime and have more variable depth/diameter ratios at a given size even when fresh. Consideration of the topography of individual ≥10 km craters thus needs to be made on a case-by-case basis during landing site characterization.

Technical Notes

For individual site characterization, a high quality suite of topographic observations exists for the Moon's surface, from Lunar Orbiter Laser Altimeter, Kaguya Terrain Camera, Lunar Reconnaissance Orbiter Camera, and other missions and instruments. These data are available in the Planetary Data System (e.g., http://imbrium.mit.edu/DATA/SLDEM2015_SLOPE/) or can be visualized with ArcGIS, JMars, or Quickmap.